Ismail Ben seddik

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Education

Indiana University Bloomington, Master of Science in Computer Science — Aug 2023 – May 2025

- Fulbright Scholarship
- GPA: 3.93/4.0
- **Relevant Coursework:** Elements of AI, Applied Machine Learning, Deep Learning Systems, Engineering Cloud Computing, Foundations of Large Language Models

Al Akhawayn University in Ifrane, Bachelor of Science in Computer Science — Sept 2018 – May 2022

- Minor: Business Administration
- GPA: 3.78/4.0
- Honors & Awards: Merit Scholarship, President's List, Dean's List, Academic Excellence Award (2019)
- **Relevant Coursework:** Linear Algebra, Calculus, Probability and Statistics, Big Data, Distributed Systems, Artificial Intelligence

Publications

EmoPAtt-Lite: Lightweight Facial Emotion Recognition

- First author. International Conference on Information Technology and Applications (ICITA 2025). Published in Springer, Lecture Notes in Networks and Systems, 2026.
- Developed a lightweight facial emotion recognition architecture incorporating attention mechanisms, achieving 79.35% accuracy on FER2013 with 1.3 million parameters.
- DOI: `https://doi.org/10.1007/978-3-032-21174-3_10`

PODSAGE: An Evidence-Grounded Multi-Agent Framework for Adaptive Learning Through Stories and Conversational Podcasts

- Presented at SAMYRAD 2026, Seville, Spain. Forthcoming in IEEE-indexed conference proceedings.
- Designed an evidence-grounded, stateful multi-agent framework that integrates learner modeling, educational content generation, assessment, and recommendation to support adaptive learning through narratives and conversational podcasts.

Conference Presentations

Response-Aware Agronomic Representation Learning for Similarity-Based NPK Recommendation

- Accepted poster presentation, 6th European Sustainable Phosphorus Conference (ESPC6), 2026.
- Developed a representation-learning and similarity-based retrieval framework integrating remote sensing, soil characteristics, and historical fertilizer-response trials to investigate uncertainty-aware nutrient recommendations in data-scarce environments.

Research Experience

Research Engineer — Data Analyst, Bioinformatics Lab, Mohammed VI Polytechnic University (UM6P) – Benguerir, Morocco — Nov 2025 – Present

- Investigating transfer learning and partial domain adaptation for field-scale crop yield prediction under data scarcity and distribution shifts.
- Developing representation-learning approaches integrating agronomic, environmental, and remote-sensing data for predictive modeling and fertilizer recommendation.
- Designing reproducible ML pipelines using MLflow, Airflow, MinIO, and Hive to support multimodal research and experimentation in an OCP Nutricrops-funded initiative.

AI Engineer, Vehicle Autonomy and Intelligence Lab, Indiana University – Bloomington, IN, USA — Oct 2023 – May 2024

- Implemented state-estimation algorithms for autonomous racing vehicles, integrating information from IMU, GNSS, and LiDAR sensors.
- Worked with real-time robotic perception and sensor-fusion systems, gaining experience in multimodal estimation and perception–action pipelines.

Selected Projects

PODSAGE: Evidence-Grounded Adaptive Learning Through Stories and Conversational Podcasts — PODSAGE

- Developed a stateful multi-agent framework integrating evidence grounding, learner modeling, content generation, and assessment to deliver adaptive educational narratives and conversational podcasts.
- Designed an Evidence-Grounding Council to guide personalized learning experiences based on domain knowledge, learning science, and evolving learner needs.
- Finalist, Cheng Wu Innovation Challenge – Indiana University.
- *Python, FastAPI, LangGraph, Ollama, PyTorch, RAG, ElevenLabs, SQLite, NetworkX*

High-Fidelity Image Reconstruction from Brain Activity — neural-image-reconstruction

- Implemented a generative decoding pipeline for reconstructing visual stimuli from fMRI recordings using pretrained latent diffusion models.
- Integrated VDVAE-derived latent representations with Versatile Diffusion and CLIP-based vision–language representations to support image reconstruction from brain activity.
- *Python, PyTorch, VDVAE, CLIP, Versatile Diffusion*

Computational Modeling of Human Visual Categorization — categorization-modeling

- Developed a convolutional neural network to estimate multidimensional psychological feature representations of natural rock images from human similarity judgments.
- Integrated learned representations into the Generalized Context Model to model human categorization behavior, connecting deep visual representations with cognitive theories of classification.
- *Python, TensorFlow, Keras, scikit-learn*

Posner Spatial Cueing — Visual Attention Experiment — posner-cueing

- Implemented a computerized Posner spatial cueing task using PsychoPy to study spatial attention and behavioral responses to valid and invalid cues.
- Developed trial presentation and response collection procedures to capture reaction time and accuracy in spatial target localization.
- *Python, PsychoPy*

Fast Periodic Visual Stimulation (FPVS) — Visual Perception Experiment — fpvs

- Developed a browser-based visual stimulation application using Vue 3 and Pinia for fixed-rate, display-aligned stimulus presentation.
- Implemented per-stimulus browser-level onset estimation, time- and frequency-domain timing diagnostics, and raw experimental data export.
- *JavaScript, Vue 3, Pinia*

Curriculum Learning for Natural Language Understanding — curriculum-learning-for-NLU

- Reproduced a published curriculum-learning approach for BERT-based natural language understanding, investigating the effects of training-example ordering on model learning.
- *Python, PyTorch, BERT*

Teaching & Mentorship

Associate Instructor, Indiana University – Bloomington, IN, USA — Aug 2024 – May 2025

- Taught laboratory and classroom sessions in Pervasive Computing and Modeling & Simulation.

- Developed instructional materials and mentored students in mobile computing, embedded systems, and computational modeling.
- Tools: Arduino, C++, React Native, Flutter, Firebase, LAMMPS, GROMACS, OVITO.

Computer Science Instructor, Coding Mind Academy – Remote, USA Jun 2024 – Aug 2025

- Taught computer science, artificial intelligence, and American Computer Science League (ACSL) curricula to students with varied educational backgrounds.
- Designed instructional content and learning activities tailored to different levels of programming and computational proficiency.

Founder & Lead Instructor, CS Academy – Casablanca, Morocco Oct 2022 – Present

- Designing and delivering computer science curricula, exploring instructional approaches informed by learning-science principles.

Chemistry & Physics Teaching Assistant, Al Akhawayn University – Ifrane, Morocco Jan 2020 – May 2022

- Facilitated laboratory sessions, supported student learning, and assisted with course assessment.

Industry Experience

Software Engineer, Hightech Payment Systems – Casablanca, Morocco Sept 2022 – Aug 2023

- Developed and maintained payment-processing software for enterprise financial networks using C, PL/SQL, Oracle, Unix, and Jenkins.
- Investigated critical software failures and contributed to system reliability and fault-tolerant application design.

Software Engineer Intern, Renault Group – Casablanca, Morocco Feb 2022 – May 2022

- Designed and built a cross-platform digital wallet application using React Native, Spring Boot, and PostgreSQL.
- Worked with containerized deployment and CI/CD workflows using Docker, Kubernetes, GitLab CI/CD, and Google Cloud Platform.

Technical Skills

Programming: Python, C, C++, Java, JavaScript, PL/SQL, MATLAB

Machine Learning & AI: PyTorch, TensorFlow, Keras, scikit-learn, JAX, Transformers, CLIP, VDVAE, Diffusion Models, LangGraph

Scientific Computing & Perception: NumPy, Pandas, SciPy, Matplotlib, PsychoPy, OpenCV, ROS, IMU, GNSS, LiDAR

Cloud, Data & Infrastructure: Docker, Kubernetes, AWS, GCP, Azure, MLflow, Airflow, MinIO, Hive, PostgreSQL, MongoDB, SQLite, Oracle, Linux, Git

Software Engineering & Frameworks: FastAPI, Flask, Django, Spring Boot, React, React Native, Flutter, Vue.js, Node.js

Leadership & Service

Vice President of the Fulbright Association, Indiana University Apr 2024 – May 2025

Head of the Computer Science Committee, Arrachad Association in Meknes Jan 2022 – Aug 2023

Software Engineer, Al Akhawayn University's Mechatronics Team Dec 2019 – Mar 2020

Languages

Arabic (Native) • English (Fluent) • French (Fluent) • Spanish (Beginner)